Mehdi Eskandari-Ghadi

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EDUCATION

Civil Engineering Ph.D. (GPA 4.0) — **Boulder, CO**
Engineering Science Focus — 2016 – 2021
University of Colorado Boulder

Thesis: "Effect of surface forces on the mechanics of adsorption-deformation in microporous media and in subcritical crack growth in reactive environment"
Advisor: Yida Zhang, Ph.D.
Area of study: Surface poromechanics, adsorption-deformation, fracture mechanics

Bachelor of Science — Civil Engineering (GPA 18.41/20) — **Tehran, Iran**
University of Tehran — 2012 – 2016
Ranked 5th among 113 undergraduate civil engineering students of the same entrance
Average GPA of all students at the Civil Engineering Department: 15.02/20

High School — Tehran, Iran
Allameh Helli 3 (NODET) High School — 2009 – 2012
Total GPA: 19.98/20

SELECTED COURSEWORK

Continuum Mechanics	A
Computational Fluid Dynamics (CFD)	A
Finite Element Method (FEM)	A
Partial Differential & Integral Equations	A
Wave-based Methods and Applications	A
Fourier Series & Boundary Value Problems	A
Complex Variables	A
Soil Dynamics	A
Soil Mechanics	A
Structural Analysis II	20/20
Structural Analysis I	19.14/20
Mechanics of Materials II	18/20
Mechanics of Materials I	18.5/20
Statics	19.9/20
Ordinary Differential Equations	20/20
Calculus II	20/20
Calculus I	20/20

HONORS AND CERTIFICATES

- Featured in PacifiCorp's Engineers Week spotlight, "PacifiCorp engineers are making a difference" (February 2024)
- Engineering Mechanics Institute (EMI) Conference award: Runner up in the 2021 Poromechanics Committee Student Poster Competition
- The Doctoral Assistantship for Excellence from University of Colorado, Boulder
- Honorary Admission for Graduate Study from the University of Tehran
- E.I.T. — Civil FE passed
- Ranked 183rd among approximately 250,000 participants in Iran's 2011 university entrance exam (Konkour)
- Ranked among the top 10 percent in every semester during Bachelor's study
- Top Researcher award in Allameh Helli 3 Seminar of Science and Technology
- Member of the Secretariat Committee of the Allameh Helli 3 Seminar of Science and Technology
- Certificate of Participation in RoboCup IRAN OPEN 2010 Competitions and Symposium
- Certificate of Education in elementary MATLAB programming
- TOEFL: 115/120 (Reading: 29, Listening: 30, Speaking: 28, Writing: 28)

- GRE: Verbal 155/170, Quantitative 170/170, Analytical Writing 4.0/6.0

PROFESSIONAL CERTIFICATIONS

Deep Dive: Data Analysis in Contour **Oct 2026**
Palantir Technologies

Speedrun: Data Science Fundamentals **Oct 2026**
Palantir Technologies

Speedrun: Your First AIP Workflow **Oct 2026**
Palantir Technologies

Foundry & AIP Builder Foundations **Sep 2026**
Palantir Technologies

Speedrun: Your First End-to-End Workflow **Sep 2026**
Palantir Technologies

ENGINEERING EXPERIENCE

Reliability & Standards Software Engineer II **Jan 2025 – Present**
PacifiCorp

- Designed, built, and deployed the company's Foundry-based facility rating data management system end to end with full-stack TypeScript and Python, migrating prototype FERC-881 line-rating and FAC-008 equipment-rating business logic into production as an integrated system. The system spans backend and queue-based processing pipelines, user-facing applications and data management UIs, audit preparation, and interdepartmental data collection, delivered concurrently with testing and production adoption.
- Built a Foundry-based ecosystem for PacifiCorp, spanning an Ontology with 100+ Actions and Functions and 30+ Pipeline Builder pipelines, to manage transmission facility and equipment records, track and approve record modifications, and compute granular, operations-scale rating calculations, including external system interfacing and efficient data management workflows delivered through automations, map interfaces, graphs, and other data solution UI elements. Owned end-to-end UX design in Foundry Workshop, including all interaction and workflow design, continuously identifying friction points and iterating on layout and information architecture to keep complex, multi-source engineering workflows simple and intuitive for end users.
- Redesigned the inherited prototype Ontology schema, which modeled facilities, linked substations, 14 equipment object types (including breakers, buses, conductors, switches, and transformers), conductor-linked structures, and element ratings linked to each equipment. Restructured each object type to hold only the properties that semantically belong to it, removing facility properties replicated on equipment and moving structure-owned geometry off conductors, and built pipelines and functions that give equipment access to linked data through the objects they link to. Unified substations and structures into a single facility locations object type, so facilities, conductors, and equipment link to structures and substations the same way and share location-related actions.
- Extended the Ontology with workflow object types: report request queues and report objects; workflow queues and workflow lineage objects backing the git-inspired workflow; analysis tag objects for organizing analyses; a dedicated rating update queue that gates which ratings can be updated and which are locked, and batches large system-wide updates over limited computation resources; and a generic queue object for large one-of-a-kind system changes such as migrations and data health cleanups.
- Established shared design conventions across all Workshop applications, including common layouts and color schemes and reusable components such as a map component, a complex facility interaction and viewing table, and metric cards used throughout the system.
- Drove adoption of the Foundry-based system by training PacifiCorp engineers and holding weekly feature-update meetings.
- Wrote user guides for the system and code documentation for its more complex features.
- Directed AI coding agents with repository and Foundry Ontology access by sequencing incremental implementations and large migrations to write and edit functions and entire features, including large formatted Excel report generation, large-scale dataframe operations logic, and React-based custom Workshop widgets, through a full commit, tag, and pull-request workflow. Coordinated agent-driven changes across multiple interdependent repositories that import one another as libraries, maintaining version and compatibility consistency across them.
- Built a reusable Python report-processing library supporting multiple internal and external data exports, and used it to power LLM-based reporting via Palantir AIP. The reporting pipeline traverses Foundry Ontology object sets into dataframes, formats outputs into schema-versioned, typed data structures with a matching data legend, and applies system, task, and report-format prompts to generate narrative reports on facility modification changes and their rating implications for admin-

istrative review and approval prior to publishing. The reports are in active use to focus administrative review across data spread over legacy tables, files, engineering drawings, and internal apps.
- Architected a reusable, git-inspired version-control framework for system analysis lifecycle management in Foundry: facility analyses are cloned, modified, and published in stages through department-specific review checkpoints, with automated resource tracking and validation.
- Designed backend Python data pipelines in Spark and Polars to ingest, parse, and remediate enterprise-scale, imperfect legacy data from multiple database and spreadsheet sources, automating cleanup and multi-source merging to reconcile records to production quality standards.
- Developed a reusable Python library implementing physics-based, IEEE-compliant thermal rating models, numerically solving the nonlinear ordinary-differential heat balance (solar heat gain, convective cooling, radiative heat loss, conductor resistance) through Runge-Kutta forward integration and Newton-Raphson inverse root-finding, vectorized at scale for static, dynamic, and transient line and equipment ratings within the platform.
- Designed and built a streamlined, user-friendly FAC-008 substation equipment rating tool from scratch in VBA/Excel, applying IEEE equipment-rating knowledge to compute normal and emergency ratings and rank limiting elements for breakers, switches, capacitors, reactors, buses, jumpers, and transformers from company standards and nameplate specifications.
- Applied geometric and spatial analysis methods to compute mutual-impedance candidate pairs, line-distance statistics, and line-to-line distance visualizations for parallel transmission corridors, integrated into platform UI graphs.
- Deployed AutoCAD experience and problem breakdown skills to implement AutoLISP one-line-drawing parsers for shape geometries and layer data. The parsers identify transmission facilities and equipment, deduce line segment and equipment configurations, and export structured data to reconcile database records with engineering drawings.

Transmission Engineer II **Feb 2024 – Jan 2025**
PacifiCorp

- Utilizing high-level knowledge of transmission line design to scope new transmission lines in collaboration with other engineering disciplines.
- Initiating process improvement pipelines for transmission design processes and documentation.
- Performing detail-oriented technical review of transmission standards, contributing to their development.
- Continued all duties from Transmission Engineer I position.

Transmission Engineer I **Jan 2023 – Feb 2024**
PacifiCorp

- Utilizing in-depth knowledge of PLSCADD, L-PILE, NESC, and GO-95 for design of new power transmission lines and rebuilding/maintenance of existing lines in states of UT, WY, OR, WA, and CA in collaboration with teams of engineers, managers, field crew, and legal.
- Met regularly with external customers to coordinate projects in which transmission lines were rerouted to meet their project needs, or company power line changes required encroaching on customer properties.
- Utilizing in-depth knowledge of structural engineering and geotechnical engineering for custom-engineered solutions of transmission structures and drilled pier foundations with reference to ASCE, ACI, and FHWA to meet special needs.
- Overhauled the company's transmission structure library from legacy templates into modular, component-based PLS-POLE libraries, and designed a new class of circuit-swap structures for repositioning circuits on a double-circuit extra-high-voltage line with minimal increase to structure size, vetted across engineering and field operations to meet electrical and structural requirements.
- Utilized VBA and Excel programming alongside transmission line rating engineering knowledge to design and build an auditable, automated FERC-881 steady-state line-rating analysis tool from scratch. This tool integrates PLS-CADD structure data, conductor properties, and historical weather data with CIGRE-referenced seasonal rating methodology to compute normal and emergency ratings and critical-span summaries. It streamlined grid line data extraction, reducing engineering hours threefold and costs by approx. $1.5M, and initialized the Foundry datasets that the system later grew from, for future system migration, validation, and automation.
- Employed Excel programming and data analysis skills to develop automated tools for deep foundation design, construction package generation, and transmission pole design aids.
- Utilized transmission engineering knowledge and programming experience to generate company sag-tension standards for tree-wire.
- Improved company standards and company procedures with detail-oriented reviews/QAQC.
- Designing and reviewing drawings for tubular steel poles and drilled piers to withstand vertical and lateral loads.

Self-motivated Projects July 2024 – Jan 2025

- Designed and developed a full-stack web app tool for project management, resource management, and business analysis using Flask, Django, JavaScript, HTML, SQL, and third-party API integration.

- Created software for transforming PLS-CADD transmission models into line rating analysis graph-based and tree-based models using Python, OOP, and design patterns.
- Developed a calculation tool for the bending capacity of hollow polygonal tapered poles by applying structural engineering skills and Python.

Structural Engineering/Construction Intern **May 2015 – Aug 2015**
Kayson Construction Co., Tehran, Iran

- Worked with a team of 15+ engineers, 20+ construction managers, and 50+ construction workers.
- Performed daily construction quality and safety inspections.
- Prepared 30+ quality and safety inspection reports.

AutoCADman and Structural Engineering Intern **May 2014 – Aug 2014**
Sazeh Pardazi Iran Consultant Co., Tehran, Iran

- Prepared 30+ sheets of technical drawings for concrete and steel structures, including beams, columns, foundations, retaining walls, slabs, and joint details.
- Apprenticed with a team of civil engineers and architects for design of steel and reinforced concrete beams, columns, and foundations.

Class Projects **Dec 2015 and May 2016**

- Utilized ETABS for the design of a 5-story residential reinforced concrete structure and steel structure under standard combinations of live, dead, wind, and seismic loads and verified by hand calculations.
- Utilized Plaxis and GeoStudio for the senior design project of retaining wall stability with and without nailing.

RESEARCH EXPERIENCE

Graduate Research Assistant / Ph.D. Candidate / Postdoctoral Researcher **Jun 2018 – Jan 2023**
University of Colorado Boulder, Boulder, CO

- Collaborated with a team of senior researchers at the Lawrence Berkeley National Laboratory to investigate and develop mechanistic models for subcritical crack growth and crack healing, with reference to 400+ research and technical articles.
- Investigated subcritical crack growth, crack healing, experimental apparatus, and utilization of optical birefringence techniques in fracture analysis with collaborators at LBNL.
- Utilized knowledge of fundamentals of fracture mechanics, fluid transport, and boundary element methods to develop software that couples fields of mechanics and physics.
- Investigated the theory and engineering application of multiphysical and multiscale solid-fluid interactions (e.g., adsorption, surface tension, solvation pressure) in micro/meso/macroporous solids.
- Developed, implemented, and validated a poromechanical model for adsorption-induced deformation of microporous material in MATLAB, based on thermodynamics, poromechanics, and surface mechanics.
- Employed technical writing and illustrative skills for preparation of 4 peer-reviewed publications in journals of JMPS, IJSS, and EFM.

Research and Implementation of granular media constitutive models **2016 – 2018**
University of Colorado Boulder, Boulder, CO

- Research and implementation of elasto-plastic hardening constitutive models for modeling of sands in geomechanics.
- Implemented granular material constitutive models in ABAQUS to investigate the response of dense and loose sands under triaxial loading and generate failure surface plots.
- Programmed a discrete element MATLAB code for modeling of grain structure stability and avalanche.

Design and Setup of Shaking Table **2009 – 2010**
Allameh Helli 3 High School, Tehran, Iran (Highschool sophomore project)

- Designed and built a shaking table for evaluation of the natural frequencies of continuous cantilever beams.

Design and Building of Line-Seeker Rescue Robot **2009 – 2010**
Allameh Helli 3 High School, Tehran, Iran (Highschool junior project)

- Designed and built a line-seeker rescue robot for participation in the RoboCup IRAN OPEN 2010 competitions.

Huffman Code Research and Implementation **2008 – 2009**
Allameh Helli 3 High School, Tehran, Iran (Highschool freshman project)

- Researched and implemented the Huffman Code Theory for loss-less data encoding and file compression.

PUBLICATIONS

Journal Articles

- Eskandari-Ghadi, Mehdi, and Yida Zhang. "Mechanics of shrinkage-swelling transition of microporous materials at the initial stage of adsorption." *International Journal of Solids and Structures* **222** (2021): 111041.
- Mehdi Eskandari-Ghadi and Yida Zhang. "Effect of pore size distribution on sorption-induced deformation of porous materials: a theoretical study." *International Journal of Solids and Structures* (2022), published online.
- Mehdi Eskandari-Ghadi, Deng Hang, Seiji Nakagawa, Steve Pride, Benjamin Gilbert, Yida Zhang. "Surface force and reactant transport as the central mechanisms for subcritical crack growth: a theoretical study." *Journal of Mechanics and Physics of Solids* (2022).
- Seiji Nakagawa, Yida Zhang, Mehdi Eskandari-Ghadi, Donald W. Vasco. "Corrections of Double-Torsion (DT) subcritical crack growth tests for crack profile geometry." *Engineering Fracture Mechanics* (2022).

CONFERENCE TALKS AND PRESENTATIONS

EMI Virtual Conference 2021 and EMI Conference 2022 **May 2021 and May 2022**

- Applied illustrative and verbal communication skills to deliver presentations titled "A multi-scale theory explaining the initial shrinkage of micro-porous solid upon gas adsorption" and "A Multiphysical Surface-Force Based Fracture Theory for Subcritical Crack Growth in Surface-Reactive Environments", followed by Q&A from engineering and academic audiences.
- Received runner up award in the 2021 Student Poster Competition for poster presentation on sorption-induced deformation of porous material.

Lawrence Berkeley National Laboratory BES — Regular Presentations **Sep 2020 – Jan 2023**

- Presented regular scientific presentations on topic of crack propagation to teams of research professionals and senior scientists in the LBNL BES with diverse backgrounds in mechanics, physics, and chemistry.
- Presented a 30-minute abstract for senior scientists in the LBNL BES annual meeting on the effects of surface forces in subcritical crack propagation.

TEACHING EXPERIENCE

Advanced Mechanics of Materials (Graduate level) — Teaching Assistant **Fall 2016, Fall 2017**
University of Colorado Boulder, Boulder, CO

- Responsible for holding office hours, preparation of homework and exam questions, and grading of homework, lab reports, and exams.
- Held 9 laboratory teaching sessions for measurement of elastic properties of materials using extension, 3-point bending, and torsion experiments.

Analytical Mechanics II — Teaching Assistant **Spring 2017**
University of Colorado Boulder, Boulder, CO

- Responsible for holding office hours, preparation of homework and exam questions, and grading of homework.

Dynamics (Undergraduate level) — Teaching Assistant **2018 – 2019**
University of Colorado Boulder, Boulder, CO

- Presented 3 class lectures, held 190 hours of office hours and exam preparation sessions, prepared and graded 6 exams and 45 homework question sets.
- Tailored technical communication specific to each student's understanding and utilized AutoCAD 3D drawings and cross sections to enhance visualization and understanding.

FE Exam Tutoring

- Tutored students for FE exam with 100% success rate.

Structural Analysis II — Teaching Assistant **Fall 2014**
University of Tehran, Tehran, Iran

- Responsible for grading homework and class projects.

Hydraulics of Open Channels — Teaching Assistant **Fall 2015**
University of Tehran, Tehran, Iran

- Responsible for holding office hours, preparation of homework and quizzes, and grading of homework and quizzes.

Math & Science Teacher **2014 – 2015**
Valiasr English Elementary School and Mesbah English Elementary School, Tehran, Iran

Private Tutoring — Mathematics and Physics **Summer 2012 and 2012 – 2014**

- Private lectures on freshman-level calculus and physics for high school and university students.

Lecture on Successful University Admission **Summer 2012**
Erfan High School and Beheshti High School, Tehran, Iran

- Delivered talks on pursuit of successful education and strategies for the National University Qualification Exam (Konkour) for women's high school audiences.

LEADERSHIP ROLES

Research Mentor **Jan 2022 – Jan 2023**
University of Colorado Boulder

- Trained new Ph.D. students on fundamentals of mechanics, written and oral skills, and technical research.
- Guided a research trainee through the modeling and simulation of beam bending in humidity actuators by application of the adsorptive poroelastic model.

President/Co-president **Aug 2018 – May 2019**
Persian Student Organization, Boulder, CO

- Proposed, secured funding, planned, and oversaw student events in collaboration with professionals from industry and academia, local businesses, and non-profit organizations.
- Proposed event plans and secured event funding with the University of Colorado Boulder Events Board.
- Managed teams of 5–10 members for event planning, advertisement, and execution.

TECHNICAL SKILLS

Programming & Software Development
Python, TypeScript/JavaScript, MATLAB, VBA, AutoLISP, FORTRAN, C++, HTML/CSS, SQL, PHP, Pascal

Platforms & Frameworks
Palantir Foundry (Ontology, Pipeline Builder, Contour, Workshop, OSDK, AIP, Spark, Polars, TypeScript/Python pipelines), Flask, Django, Palantir Foundry UI

Engineering Software
PLSCADD/PLS-TOWER, L-PILE, Autodesk AutoCAD, ABAQUS, LS-DYNA, ETABS, Plaxis, GeoStudio, SAP2000, SAFE, Civil3D, WaterGEMS, SewerCAD, StormCAD

Other Tools
Git, Microsoft Office Suite, LaTeX, Inkscape, Corel Draw, OriginPro

Engineering Standards
NESC, GO-95, FERC-881

LANGUAGE SKILLS

English — Fluent/Native
Persian/Farsi — Native

OTHER INTERESTS

- I am a plant hobbyist, animal lover, dog owner, and skilled chef for humans and dogs.
- I have a great passion for programming as a hobby.
- I enjoy cycling, swimming, jogging, and playing soccer.
- I play guitar and piano (beginner).
- I regularly watch movies and listen to music.